KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru – 560 012

Telephone: 080 – 23341652, 23348849 || Email: spp@kscst.org.in
Website: www.kscst.org.in/spp.html or https://kscst.karnataka.gov.in/en

FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 50th SERIES OF STUDENT PROJECT PROGRAMME 2026 – 2027

(Celebrating the Golden Jubilee of KSCST's flagship Student Project Programme)

CALL FOR PROJECT PROPOSALS FROM FINAL-YEAR STUDENTS
B.E. | B.Arch. | M.Tech. | M.Sc. | M.Tech. / M.Sc. (Agriculture) | MBA | MCA

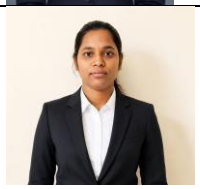
(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill-up the Google Form.)

https://forms.gle/VPPewZnbUhw1V8KdA

PART–A: GENERAL INFORMATION	
1.	**Project Title:** *EPMIC*-Breast: Predicting Breast Cancer Risk in Rural Woman Using DNA-Based Epistatic Interaction Detection and Multiple Clustering Techniques
2.	**Keywords:** Breast Cancer Prediction, DNA Data, Multiple Clustering, K-Means Clustering, Hierarchical Clustering, Non-negative Matrix Factorization (NMF), Graph-regularized NMF (GNMF), Epistatic Interaction, SNP Analysis, Feature Selection, Machine Learning, Genomics, Bioinformatics.
3.	**Broad Theme:** 1. **Theme No.:**7 2. **Theme Name:** Healthcare, Biotechnology and Biomedical Engineering 3. **Sub-theme / Focus Area:** Genomics, Bioinformatics and Precision Medicine

4.	**Institution Details:** 1. **Name of Institution:** S J C INSTITUTE OF TECHNOLOGY 2. **Course Name:** MAJOR PROJECT PHASE 2 3. **Department Name:** ARTIFICIAL INTELLIGENCE AND DATA SCIENCE 4. **Affiliated University:** VTU 5. **District of Your Institution:** CHICKBALLAPUR 6. **Complete Postal Address of the Institution with Pin code:** SJC INSTITUTE OF TECHNOLOGY,P.B.NO:20,B.B.ROAD,CHICKBALLAPUR-562101,KARNATAKA
5.	**Project Guide Details:** 1. **Project Guide Name:** Mr. BADRINATH K, Dr. SHWETHA GR 2. **Project Guide Designation:** ASSISTANT PROFESSOR, ASSOCIATE PROFESSOR 3. **Project Guide Department:** ARTIFICIAL INTELLIGENCE AND DATA SCIENCE 4. **Project Guide Email id:** badrinathk@sjcit.ac.in 5. **Project Guide Mobile Number:** 9743625894
6.	**Student Team Details:**

Sl. No.	Insert Passport-size Photograph	Name of the Student (BLOCK LETTERS)	USN / Registration No.	Degree & Branch & Semester	Category as per College Records (SC/ST/ OBC/ General)	Mobile No.	Email Id
1.		Ms. SETTIVARI BHAVYA SREE	1SJ23AD051	**Degree:** BE **Branch:**AI&DS **Semester:** VII	GENERAL	7569558150	bhavyasreeroyal2005@gmail.com

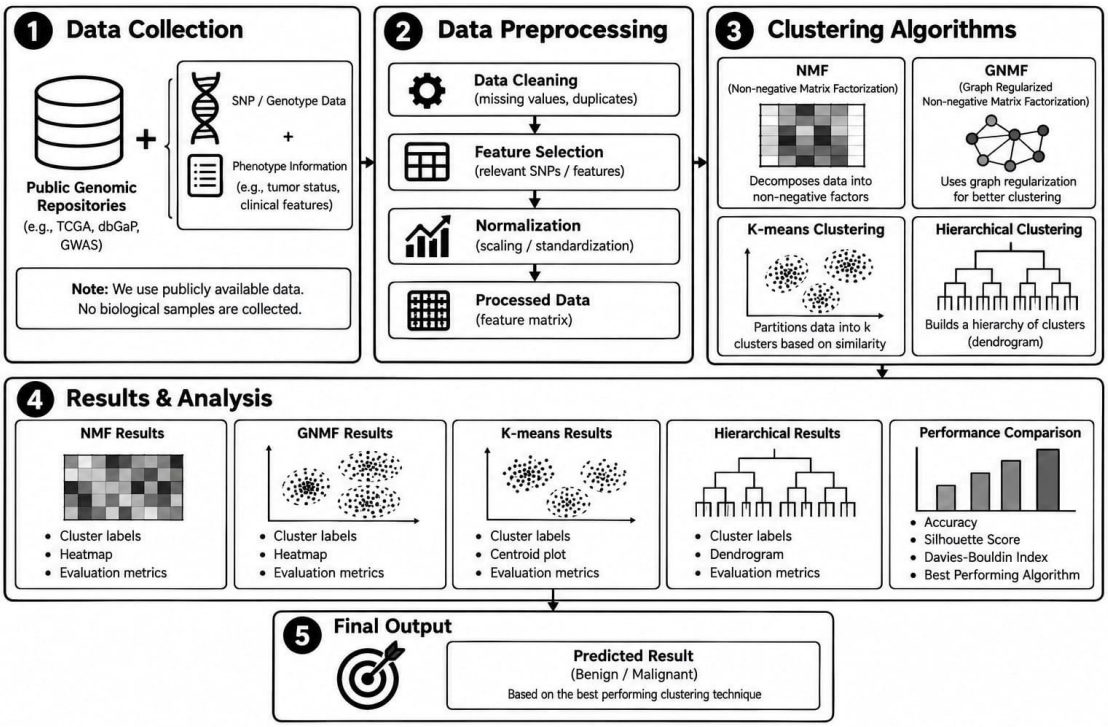
2.		Ms. KONDETI CHANDRA PRIYA	1SJ23AD021	**Degree:** BE **Branch:**AI&DS **Semester:** VII	GENERAL	9353969108	Kondetichandrapriya @gmail.com
3.		Ms. MEDIMI GAGANA SREE	1SJ23AD028	**Degree:** BE **Branch:**AI&DS **Semester:** VII	GENERAL	7842883214	gaganasreemedimi2005 @gmail.com
4.		Ms. PALLAVI B GOWDA	1SJ23AD035	**Degree:** BE **Branch:**AI&DS **Semester:** VII	OBC	9663978936	Pallavibgowda3 @gmail.com

7. **SPP Coordinator Details:**

1. **SPP Coordinator Name:** Mrs. E.N.SRIVANI
2. **SPP Coordinator Designation:** ASSISTANT PROFESSOR
3. **SPP Coordinator Email:** srivani1598.ece@sjcit.ac.in
4. **SPP Coordinator Mobile Number:**8639902870

PART - B: PROJECT DESCRIPTION	
8.	**Background and Need of the Project:** Breast cancer is one of the major health concerns worldwide, and identifying genetic patterns associated with the disease can support research and early risk assessment. DNA contains genetic variations such as Single Nucleotide Polymorphisms (SNPs) that may be associated with an individual's susceptibility to certain diseases. In the existing scenario, breast cancer assessment mainly uses clinical examination, family history, imaging techniques, and genetic testing, while analyzing large DNA datasets remains challenging because they contain many complex genetic features. The problem addressed by our project, "Predicting Breast Cancer Using Multiple Clusterings Through DNA," is to analyze DNA/SNP data and identify meaningful patterns among individuals that may be associated with breast cancer. The project applies multiple clustering techniques such as Non-negative Matrix Factorization (NMF), Graph-based NMF (GNMF), and Hierarchical K-Means to group similar genetic patterns and discover hidden relationships in the data. The need for the project arises from the difficulty of manually analyzing high-dimensional genetic data and the limitations of using a single clustering technique. By combining multiple clustering approaches, the project aims to organize complex DNA data, identify useful genetic patterns, and support further breast cancer research. The beneficiaries of this project include researchers, bio informatics professionals, healthcare researchers, medical institutions, and students, who can use the approach for studying genetic patterns and developing further research-based applications. The proposed system is intended as a research and analytical tool and not as a clinical diagnostic system.
9.	**Review of Existing Work:** Existing work on breast-cancer risk assessment uses genetic testing, SNP analysis, machine-learning techniques, gene-expression analysis, and clustering methods to identify genetic patterns associated with breast cancer. Research has explored the use of multiple SNPs for estimating breast-cancer risk, while computational methods such as NMF and graph-based approaches have been applied to discover patterns in high-dimensional biological data. Commercial solutions such as My Risk Hereditary Cancer Test analyze 63 genes associated with hereditary cancers and can combine genetic and clinical information for breast-cancer risk assessment. Several patents have also proposed assessing breast-cancer risk using combinations of SNPs and genetic information. However, existing solutions generally focus on specific genes, selected SNPs, risk scores, or particular analytical methods. This creates a research opportunity to explore multiple clustering techniques together for analyzing complex DNA/SNP datasets. Therefore, our project, "Predicting Breast Cancer Using Multiple Clusterings Through DNA," proposes the use of NMF, GNMF, and Hierarchical K-Means to identify and compare different genetic patterns in the dataset. The project is intended as a research and analytical framework, rather than a clinical diagnostic system.

10.	**Problem Statement:** Breast cancer is associated with various genetic factors, and DNA/SNP data contains complex patterns that are difficult to identify using conventional methods. Existing approaches often focus on individual genes, selected SNPs, or a single analysis technique. Therefore, there is a need for an effective computational approach to analyze DNA/SNP data and identify meaningful genetic patterns associated with breast cancer. This project addresses the problem by applying multiple clustering techniques—NMF, GNMF, and Hierarchical K-Means—to group similar genetic patterns, discover hidden relationships, and support breast-cancer research.
11.	**Objectives:** • To collect and preprocess DNA/SNP data by converting genetic information into a suitable numerical format for efficient analysis. • To identify important SNPs and genetic patterns associated with breast cancer from high-dimensional genetic data, thereby reducing the search space and data complexity. • To apply multiple clustering techniques—NMF, GNMF, and Hierarchical K-Means to discover and group similar genetic patterns and improve pattern identification. • To compare and evaluate the clustering results using suitable performance measures and visualizations to identify meaningful breast-cancer-associated patterns while reducing computational time and complexity.
12.	**Innovation / Novelty:** The proposed project introduces an integrated framework for identifying breast-cancer-associated genetic patterns from DNA/SNP data by combining NMF, GNMF, and Hierarchical K-Means. NMF extracts important hidden genetic patterns from high-dimensional SNP data, while GNMF considers the similarity between genetically related samples. Hierarchical K-Means further groups the samples into meaningful clusters at different levels. The novelty lies in combining these three complementary techniques and comparing their clustering results to identify consistent and meaningful genetic patterns. This approach can reduce data complexity, reveal hidden relationships among genetic samples, and provide an interpretable computational framework to support further breast-cancer genomic research.
13.	**Proposed Methodology:** • **Data Collection:** Collect DNA/SNP data containing genetic variations and breast-cancer status from a suitable dataset. • **Data Preprocessing:** Clean the data, handle missing values, and convert DNA/SNP information into numerical values suitable for analysis. • **Feature Preparation:** Organize the SNP features and samples into a structured data matrix for clustering. • **NMF:** Apply Non-negative Matrix Factorization to reduce data complexity and extract important genetic patterns. • **GNMF:** Apply Graph-based NMF to consider similarities between samples and identify closely related genetic patterns.



- **Hierarchical K-Means:** Apply Hierarchical K-Means to group samples according to their genetic similarities.
- **Cluster Comparison:** Compare the clusters produced by all three methods to identify common and distinct genetic patterns.
- **Pattern Analysis:** Analyze the resulting groups to determine patterns associated with breast-cancer samples.
- **Evaluation:** Evaluate and visualize the clustering results using suitable clustering measures.
- **Output:** Present the identified genetic clusters and patterns through tables, graphs, and visualizations for further research.

14. Expected Deliverables:

- **Working Software Prototype –** A functional system for analyzing DNA/SNP data.
- **Preprocessed DNA/SNP Dataset** – Cleaned and numerical genetic data ready for analysis.
- **Multiple Clustering Module –** Implementation of NMF, GNMF, and Hierarchical K-Means.
- **Cluster Analysis Results –** Identified genetic groups and patterns associated with breast cancer.
- **Visualization Dashboard –** Graphs and charts showing clustering results and genetic patterns.
- **Comparison Report –** Comparison of the performance and results of the three clustering techniques.
- **Research Report –** Complete documentation of methodology, results, findings, and limitations.

15. **Expected Outcomes and Impact:** The project is expected to provide a DNA/SNP-based framework for identifying meaningful genetic patterns associated with breast cancer using NMF, GNMF, and Hierarchical K-Means. It will simplify the analysis of complex genetic data, compare clustering results, and generate useful patterns for further research. The system can benefit researchers, bioinformatics professionals, healthcare institutions, and students by providing a practical and cost-effective analytical tool for studying breast-cancer-related genetic patterns. It can also support future development of advanced genetic risk-analysis systems, but is intended for research purposes and not direct clinical diagnosis.

16. **Relevance:**

Tick wherever applicable

- □ **SDGs (Sustainable Development Goals)**
- □ **Karnataka Priority Sector**
- □ **Rural Development**
- □ **MSME**
- □ **Agriculture**
- ☑ **Healthcare**
- □ **Water**
- □ **Environment**
- □ **Climate**
- □ **Manufacturing**
- □ **Smart Cities**
- □ **Digital India**
- □ **Startup**
- □ **Women Empowerment**
- □ **Others (Please Specify):**

17 . **Work Plan (Please specify the month and tick ✓)**

1.Laboratory	Computer Lab
2.Equipment	Computers
3.Software	Python
4.Instruments	None
5.Workshop	None
6.Institutional facilities	Internet & Library

18 **Facilities Available:**

Activity	Month 1	Month 2	Month 3	Month 4	Month 5
Literature Survey	✓	✓			
Design		✓	✓		
Development			✓	✓	
Testing				✓	✓
Validation					✓
Final Report					✓

19. Major Requirements for the Project:

Hardware

- Computer/Laptop
- Internet Connection
- Basic Storage

Software

- Python
- VS Code
- NumPy
- Pandas
- Scikit-learn
- Matplotlib

Consumables

- None

Field Studies

- Biological Samples (if using an existing/public DNA-SNP dataset)

20. Budget Estimate (Remove whichever is not relevant for the project):

Item	Estimated Cost (₹)
Equipment / Components	2000
Consumables	500
Fabrication	0
Software / Cloud Services	1,000
Field Survey / Data Collection	2,000
Testing / Analysis	3,500
Miscellaneous	1,000
Total	10,000

21. References:

[1] Journal Paper – DNA + Machine Learning + ClusteringC. C. Young et al., "Development and validation of a reliable DNA copy-number-based machine learning algorithm (CopyClust) for breast cancer integrative cluster classification," Scientific Reports, vol. 14, 2024, Art. no. 11861, doi: 10.1038/s41598-024-62724-6.This is highly relevant because it uses DNA copy-number data, machine learning and breast-cancer clustering.

[2] Journal Paper – Genomics + Breast Cancer E. Ambrosino et al., "The WHO genomics program of work for equitable implementation of human genomics for global health," Nature Medicine, vol. 30, pp. 2711–2713, 2024, doi: 10.1038/s41591-024-03225-x.This provides a recent reference for the use and implementation of human genomic information in healthcare.

[3] Conference Paper – Breast Cancer + Clustering + MLC. Y. Lim et al., "Integrating Supervised and Unsupervised Learning for Prognostic Prediction in Breast Cancer," in 2025 IEEE 15th International Conference on System Engineering and Technology (ICSET), 2025, pp. 300–305, doi: 10.1109/ICSET65917.2025.11284246.

This is useful for your project because it discusses unsupervised clustering together with machine learning for breast cancer.

[4] International Standard – Genomic DNA DataISO/TS 20428:2024, Genomics Informatics — Data elements and their metadata for describing structured clinical genomic sequence information in electronic health records, International Organization for Standardization, 2024.This standard is relevant to the DNA/genomic-data representation and metadata part of your project. It specifically covers DNA-level variation from human samples using sequencing technologies.

[5] Government/International Health Report – Breast CancerWorld Health Organization (WHO), Breast Cancer, WHO Fact Sheet, 2026.This can be used for current background information, disease statistics, risk factors and the importance of early detection. WHO reports an estimated 2.4 million women diagnosed with breast cancer globally in 2024.

22. Processing Fee Payment Details:

Important Instructions:

Payment Details:

1. Mode of Payment: ✓ UPI ☐ NEFT
2. Transaction / UTR / Reference Number:029404625954
3. Date of Payment: 14/10 / 2026

4. Amount Paid: ₹1,180/-
5. Name of Account Holder: SETTIVARI BHAVYA SREE
6. Name of Bank (for NEFT):UNION BANK
7. UPI ID Used:7569558150@axl

Attachment:

✓ Copy of UPI Payment Receipt / NEFT Transaction Receipt enclosed.

23. Student Details with Signature:

Sl. No.	Name of Student (Block Letters)	USN / Registration No.	Signature	Date
1.	SETTIVARI BHAVYA SREE	1SJ23AD051		
2.	KONDETI CHANDRA PRIYA	1SJ23AD021		
3.	MEDIMI GAGANA SREE	1SJ23AD028		
4.	PALLAVI B GOWDA	1SJ23AD035		

(Name & Signature of SPP Coordinator with Seal)

Email id: E.N.SRIVANI

Contact No.:8639902870

Name and signature of the Project Guide with Seal:

Email id: badrinathk@sjcit.ac.in

Contact No.: 9743625894

Mandatory Attachments

☐ Project Proposal (PDF) duly signed by the Project Guide, Head of the Department and Principal/Head of the Institution.

☐ Declaration and Endorsement by the Head of the Institution.

☐ Copy of the Processing Fee Payment Receipt (UPI/NEFT).

☐ Any other supporting documents, if applicable.

DECLARATION

(From the Project Team)

(To be signed by all student members, scanned, and uploaded along with the project proposal)

Project Details

Title of the Project: **EPMIC-Breast: PREDICTING BREAST CANCER RISK IN RURAL WOMAN USING DNA-BASED EPISTATIC INTERACTION DETECTION AND MULTIPLE CLUSTERING TECHNIQUES**

Broad Theme: **HEALTHCARE, BIOTECHNOLOGY AND BIOMEDICAL ENGINEERING**

Sub-theme / Focus Area: **GENOMICS, BIOINFORMATICS AND PRECISION MEDICINE**

Department: **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

Institution: **S J C INSTITUTE OF TECHNOLOGY**

We, the undersigned members of the above project team, hereby declare that the information furnished in this project proposal is true, complete, and correct to the best of our knowledge and belief.

Student Details

Sl. No.	Name of Student	USN / Registration No.	Signature	Date
1.	SETTIVARI BHAVYA SREE	1SJ23AD051		
2.	KONDETI CHANDRA PRIYA	1SJ23AD021		
3.	MEDIMI GAGANA SREE	1SJ23AD028		
4.	PALLAVI B GOWDA	1SJ23AD035		

(Name & Signature of SPP Coordinator with Seal)

Email id: Srivani1598.ece@sjcit.ac.in

Contact No.: 8639902870

Name and signature of the Project Guide with Seal:

Email id: badrinathk@sjcit.ac.in

Contact No.:9743625894

ENDORSEMENT

This is to certify that:

1. Ms. SETTIVARI BHAVYA SREE
2. Ms. KONDETI CHANDRA PRIYA
3. Ms. MEDIMI GAGANA SREE
4. Ms. PALLAVI B GOWDA

are bonafide final-year students of the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE, pursuing BE at S J C INSTITUTE OF TECHNOLOGY

The institution has scrutinized and approved this project proposal for submission to the **Karnataka State Council for Science and Technology (KSCST)** under the **50th Series Student Project Programme (SPP): 2026 - 2027**.

(Name & Signature of Project Guide with Seal)	**(Name & Signature of SPP Coordinator with Seal)**	**(Signature of HoD with Seal)**
Email id: badrinathk@sjcit.ac.in	**Email id:** Srivani1598.ece@sjcit.ac.in	**Email id:** hodaids@sjcit.ac.in
Contact No.: 9743625894	**Contact No.:**8639902870	**Contact No.:** 9972925814

(Signature of the Principal with Seal)

Email id:principal@sjcit.ac.in
ContactNo.:08156-200183-184-185-187

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note:** Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348849, email: finance@kscst.org.in

1. TITLE OF THE PROJECT	:	EPMIC-Breast: PREDICTING BREAST CANCER RISK IN RURAL WOMAN USING DNA-BASED EPISTATIC INTERACTION DETECTION AND MULTIPLE CLUSTERING TECHNIQUES
2. NAME OF THE TEAM LEADER	:	SETTIVARI BHAVYA SREE
3. EMAIL ID	:	bhavyasreeroyal2005@gmail.com
4. CONTACT MOBILE NO.	:	7569558150

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	029404625954
6. TRANSACTION ID	:	T261014120100564372984 4
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	SETTIVARI BHAVYA SREE 7569558150
8. NAME OF THE BANK	:	UNION BANK OF INDIA
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	14/10/2026
11. TIME	:	12:01 PM
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	Online UPI

(Name & Signature of the team leader)

(Name & Signature of Project Guide / SPP Co-ordinator / HOD with Seal)

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

50th SERIES OF STUDENT PROJECT PROGRAMME (SPP)

(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)

BANK ACCOUNT DETAILS OF KSCST

Name and address of the Institution	Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012
Account holder's name / Designation	Secretary, Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank, IISc Campus Branch, Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank, Indian Institute of Science, Bangalore-560012

BANK DETAILS

Name of the Agency	Karnataka State Council for Science and Technology IISc Campus, Bangalore - 560012
Account holder's name / Designation	Secretary , Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank IISc Campus Branch Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank Indian Institute of Science Bangalore-560012

Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348849 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in



Note: For finance related queries, please contact KSCST accounts section: 080 – 23341652, 23348849 during office hours between 3.00 pm to 5 pm, email: finance@kscst.org.in with a copy to spp@kscst.org.in